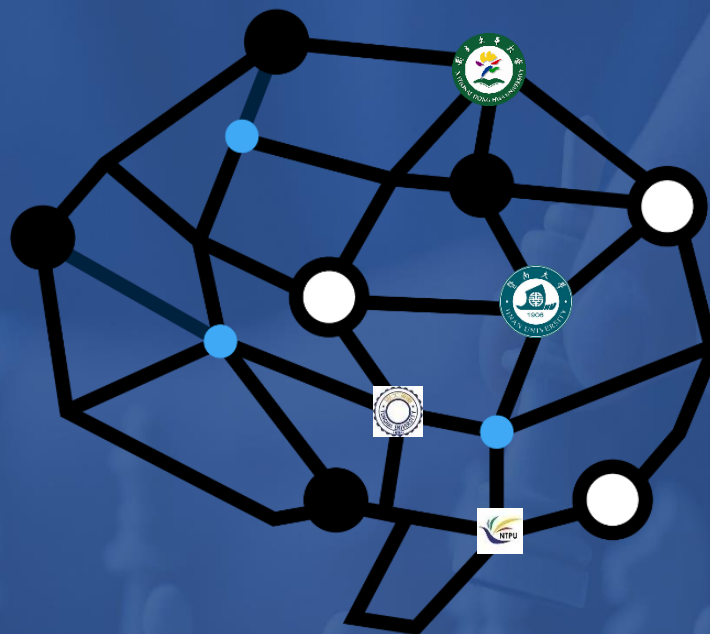


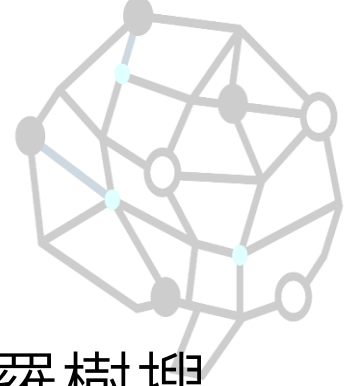


Chap. 7 AlphaGo的原理

課程：人工智慧導論

授課教師：周信宏

AlphaGo



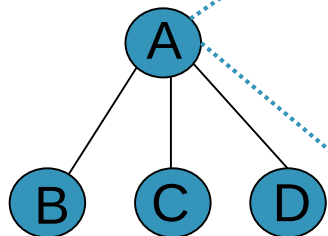
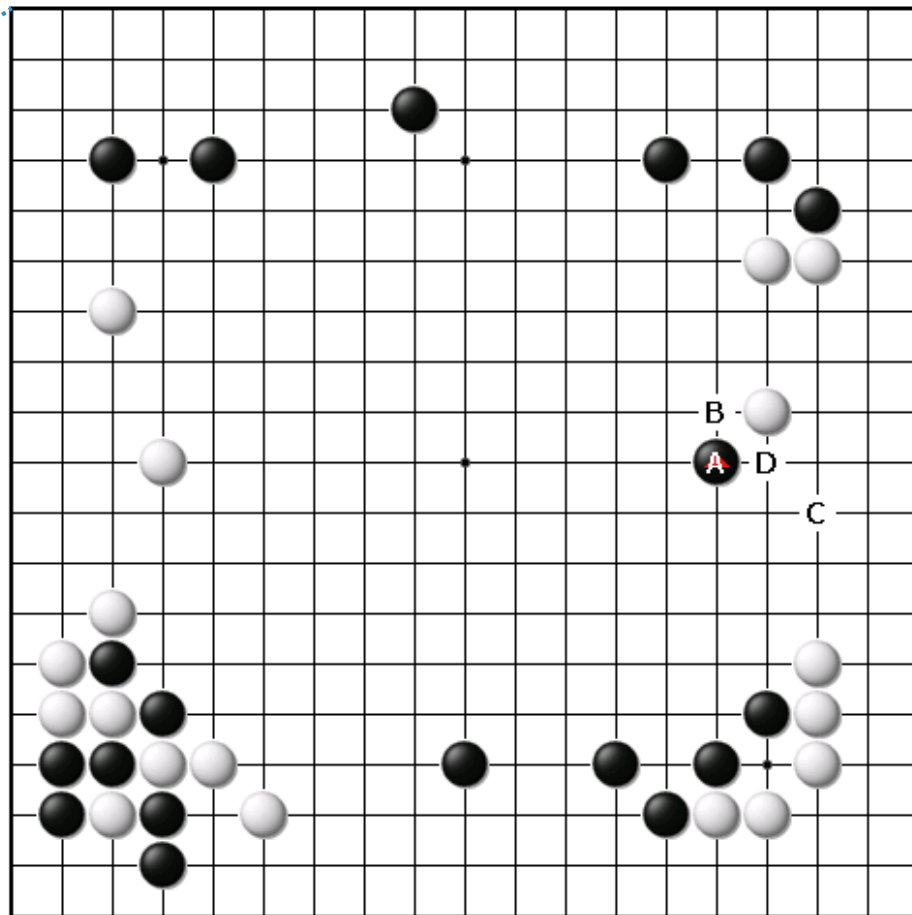
- MCTS + PN + VN + Rollout
- MCTS(Monte Carlo Tree Search, 蒙地卡羅樹搜尋)
 - 思考推理的架構
- Policy Network(策略網路)
 - 記憶學過的棋型，從中找出目前策略
- Value Network (評估網路)& Rollout policy(隨機策略)
 - 形勢判斷

樹狀搜尋範例

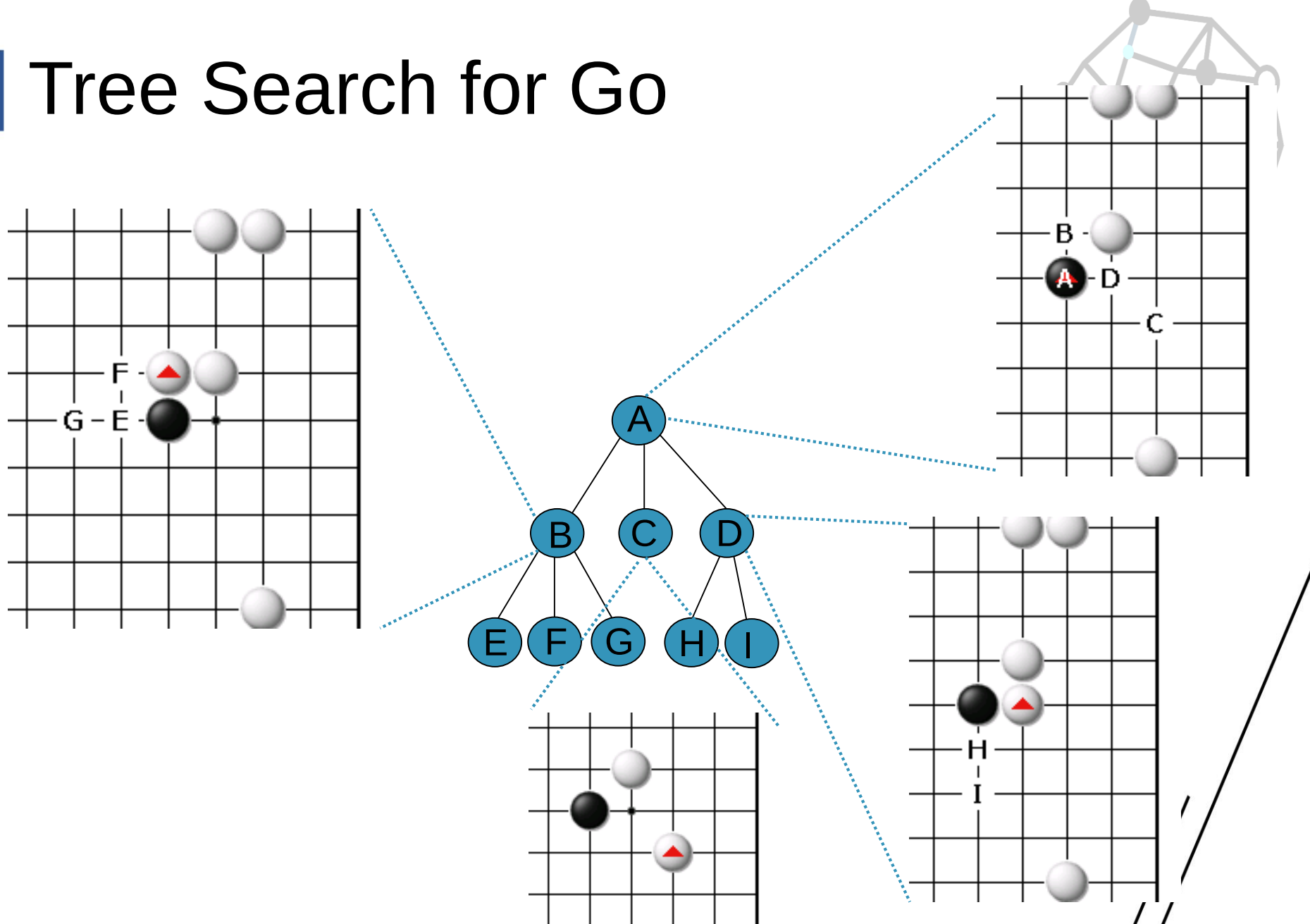


A B C D E F G H J K L M N O P Q R S T

19
18
17
16
15
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2
1



Tree Search for Go



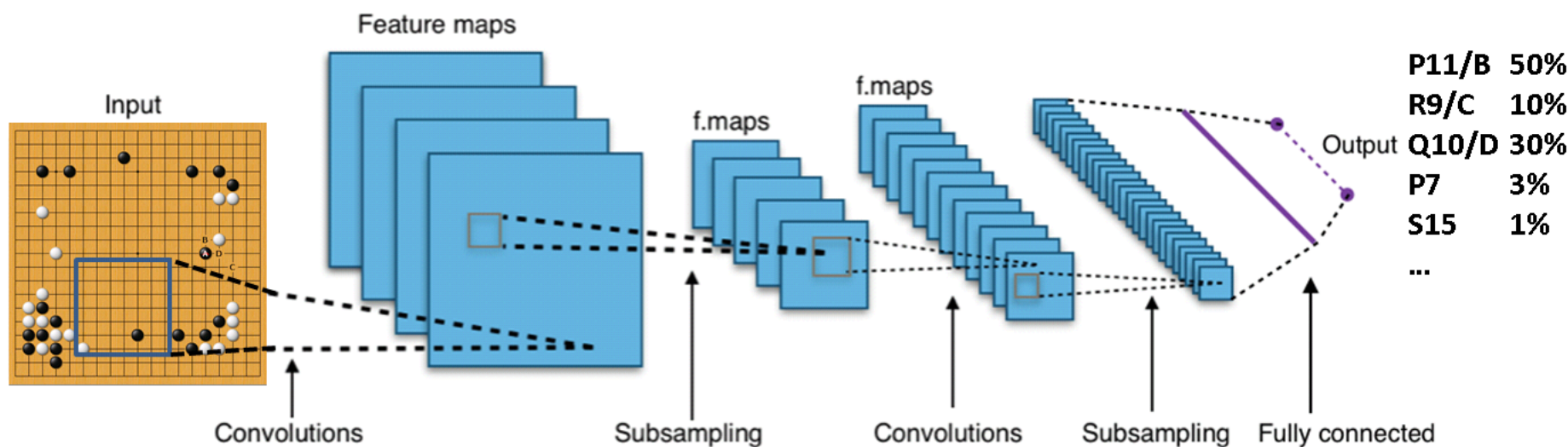
甚麼是PN, VN的N?

- Deep Learning **N**etwork(深度學習類神經網路)
- Multi-class Classifier(分類器)
 - Inputs are pixel
 - Outputs are classes possibilities
- 我想知道圖中甚麼?



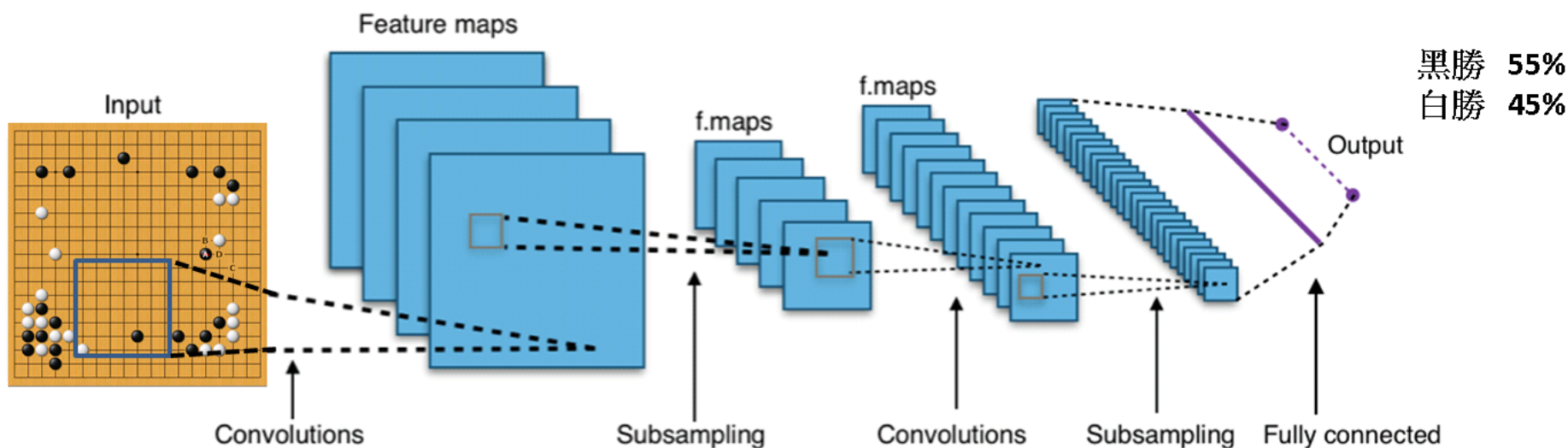
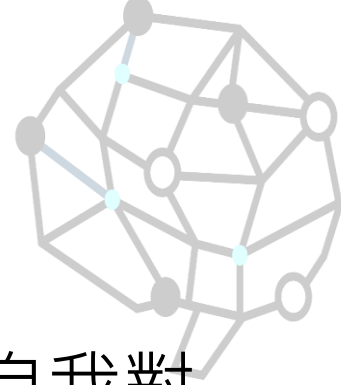
PN(策略網路)

- 我想知道那個著手是好棋
- Training data: KGS 3千萬個盤面

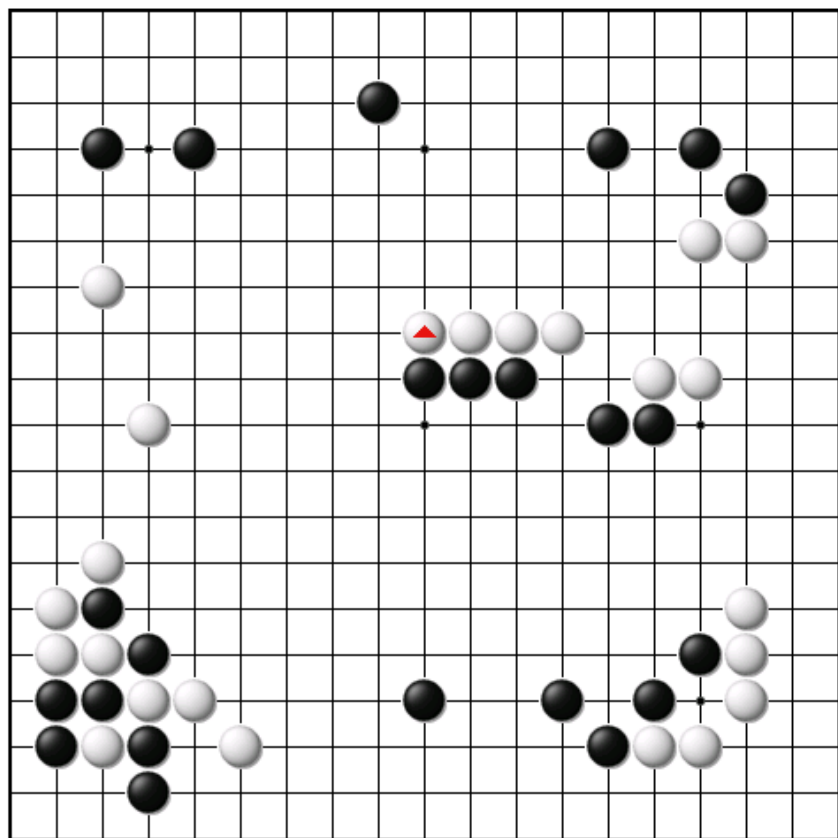


Value Network (評估網路)

- 我想做形勢判斷
- Training data: 3千萬個盤面，從3千萬盤自我對局取得



AlphaGo運作圖解



卡羅樹

最手 → 判盤

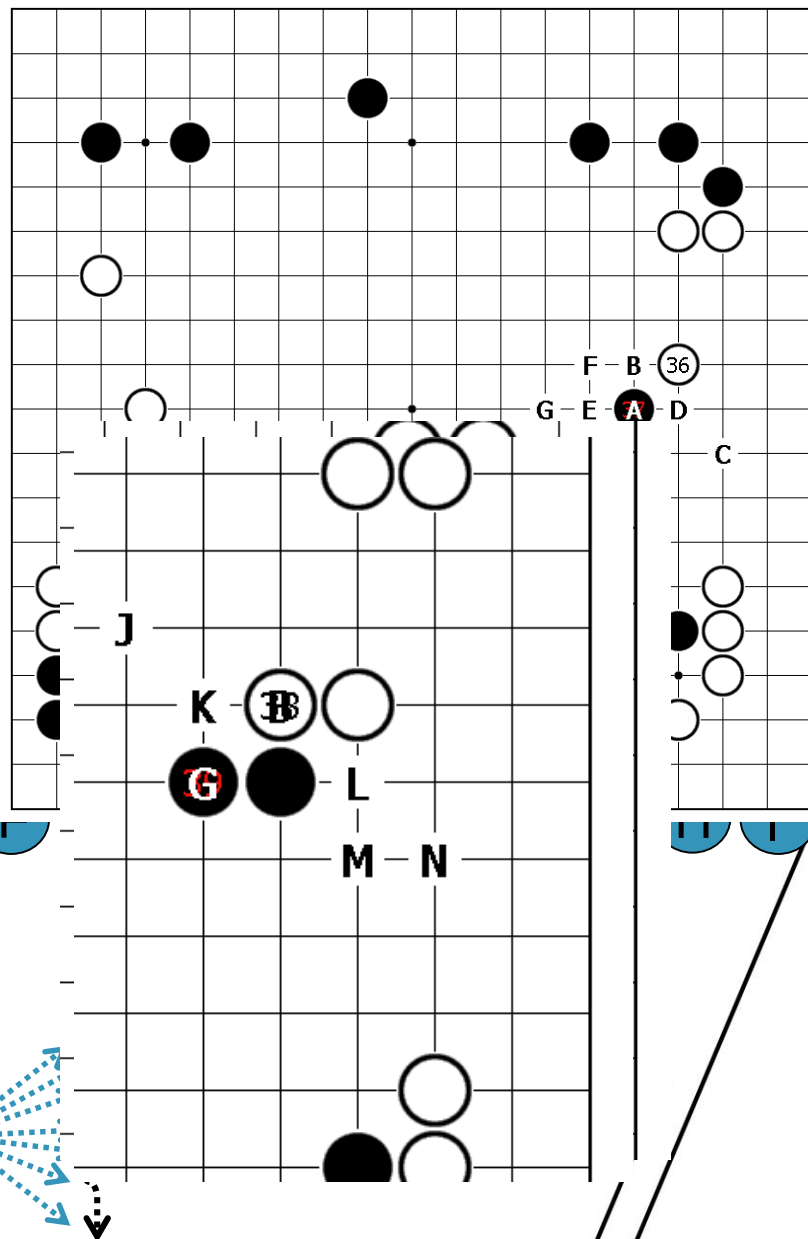
三, ABF,
ADH,

化圖時
擇其中
一手

I E

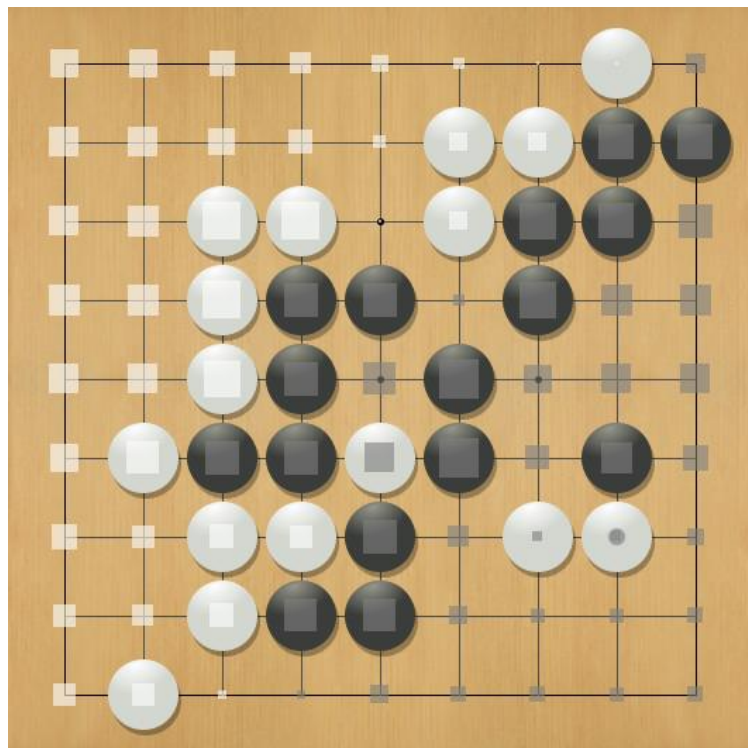
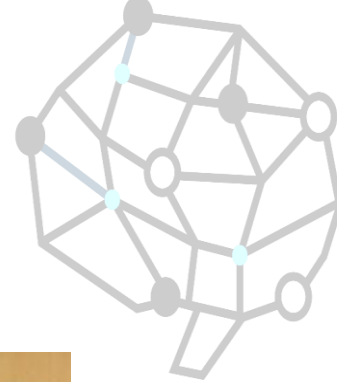
0.03

MC Rollout

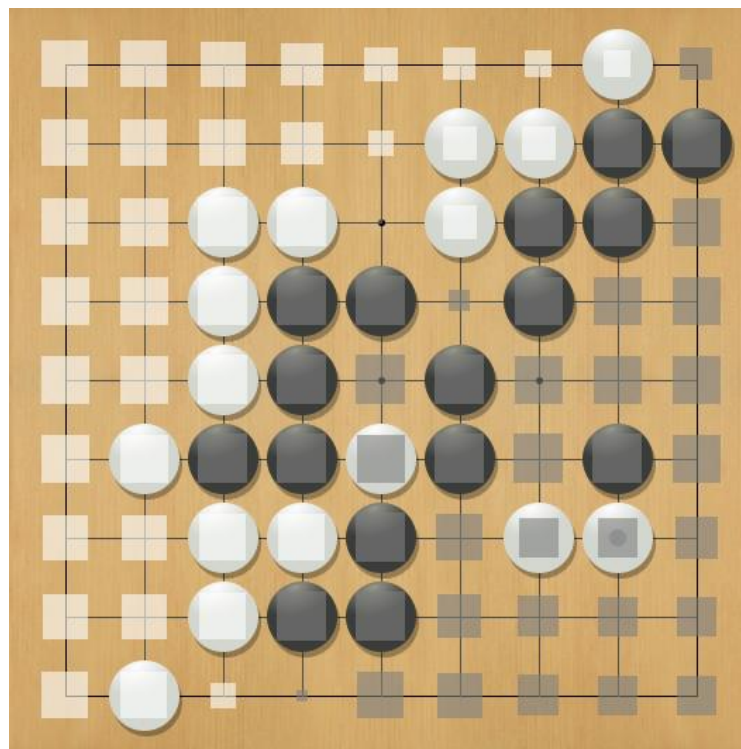


MCTS

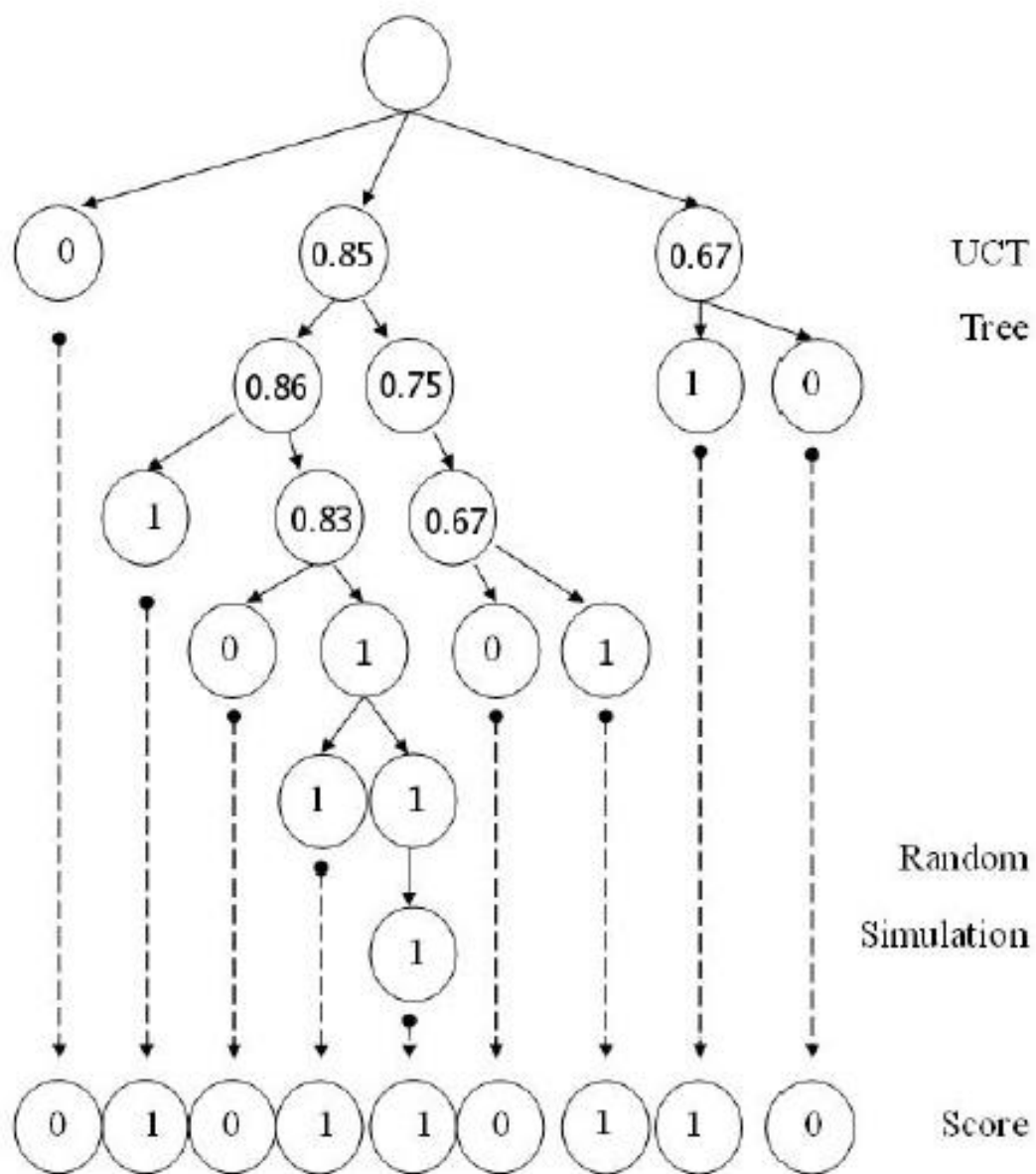
Rollout正確率的影響



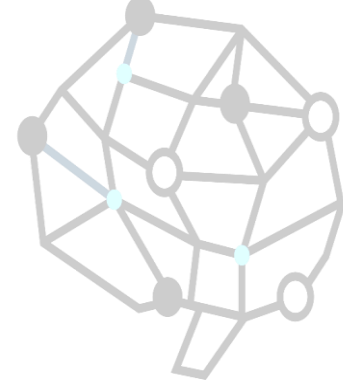
All random



Using Heuristic



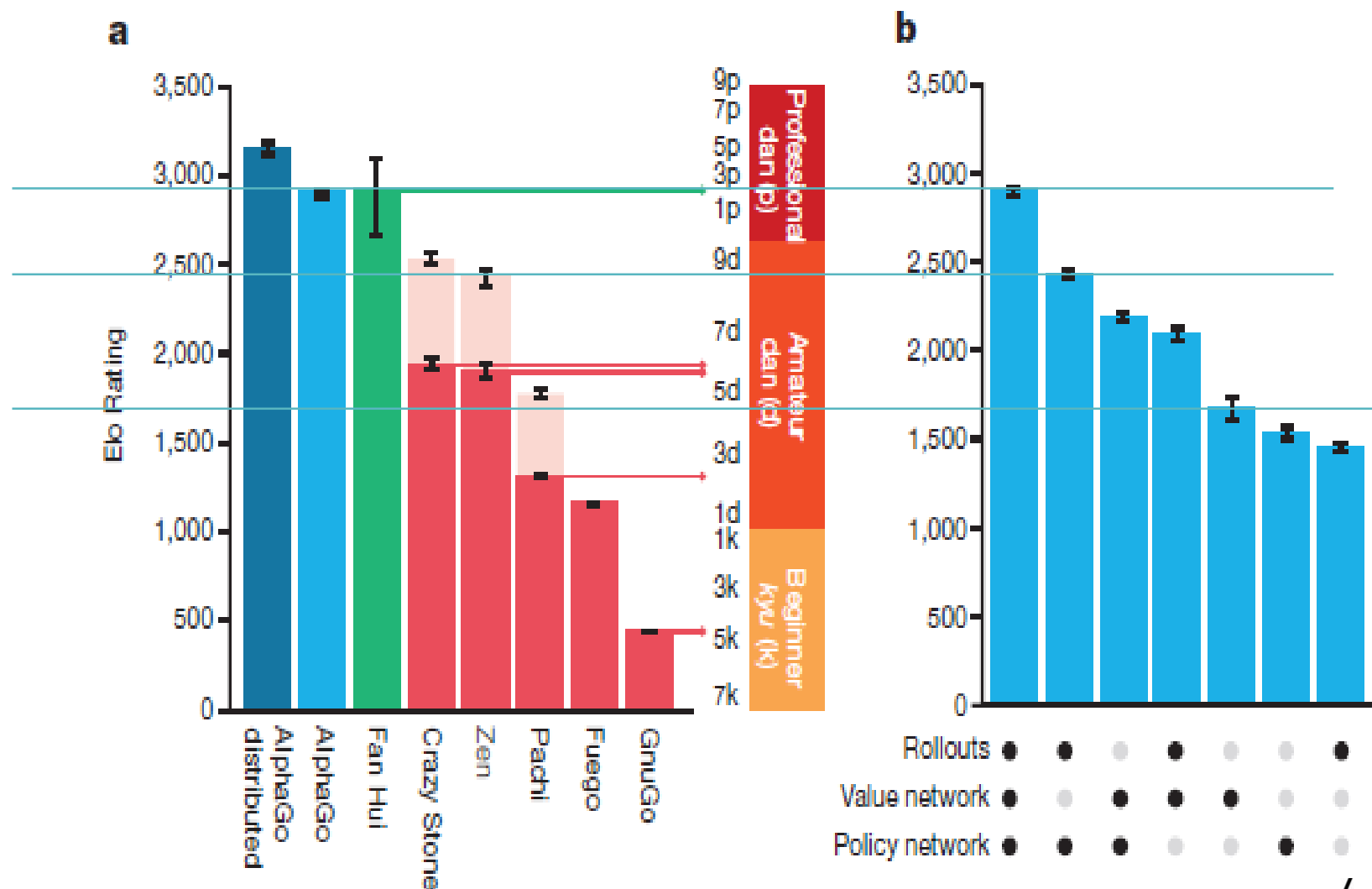
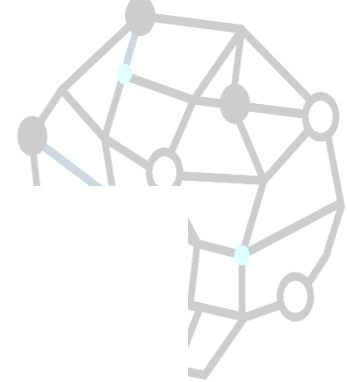
AlphaGo 組成元件時間成本



- 正確率與成本

Name	策略網路	RL policy	Rollout policy	Value Network
目的	著手預測	產生自我對局盤面	產生模擬的著手	形式判斷
正確率	57%	NA	24.2%	NA
時間成本	3ms	3ms	2us	NA

AlphaGo 與眾不同之處



AlphaGo硬體

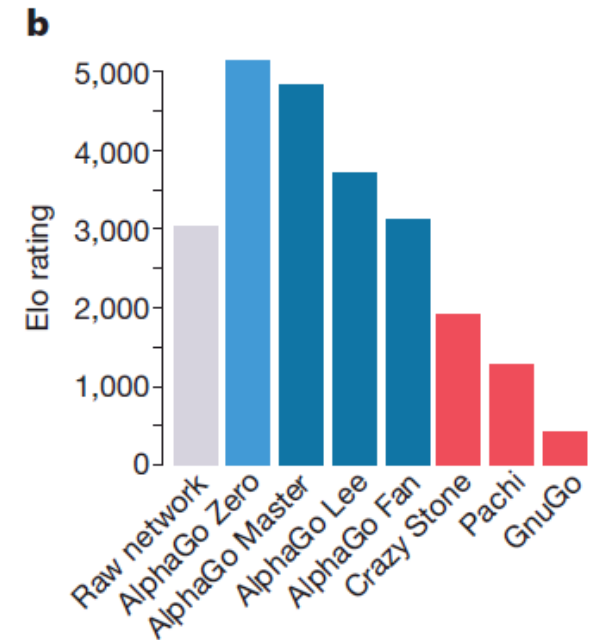
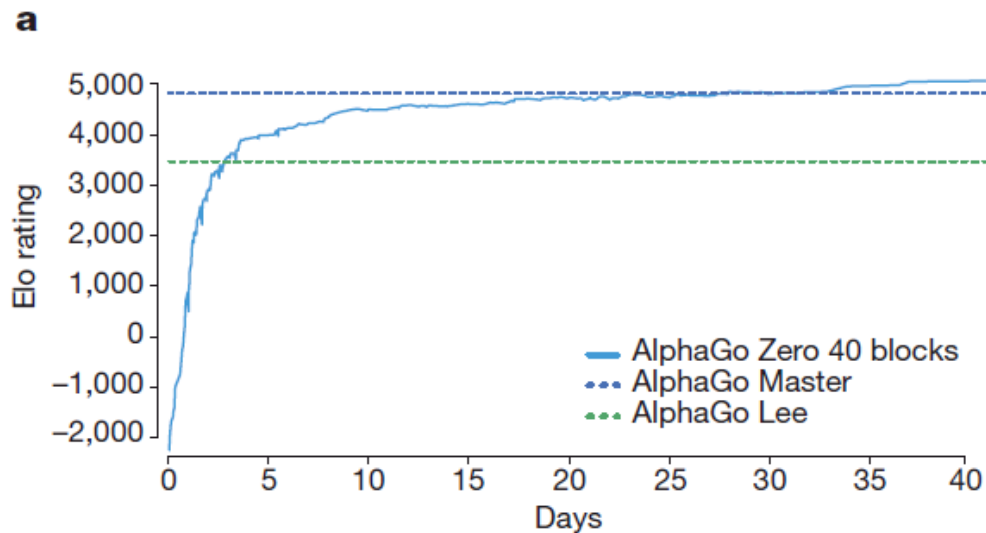


配置與效能				
配置	搜尋執行緒數	CPU核心數	GPU數	Elo等級分
單機	40	48	1	2,151
單機	40	48	2	2,738
單機	40	48	4	2,850
單機	40	48	8	2,890
分散式	12	428	64	2,937
分散式	24	764	112	3,079
分散式	40	1,202	176	3,140
分散式	64	1,920	280	3,168

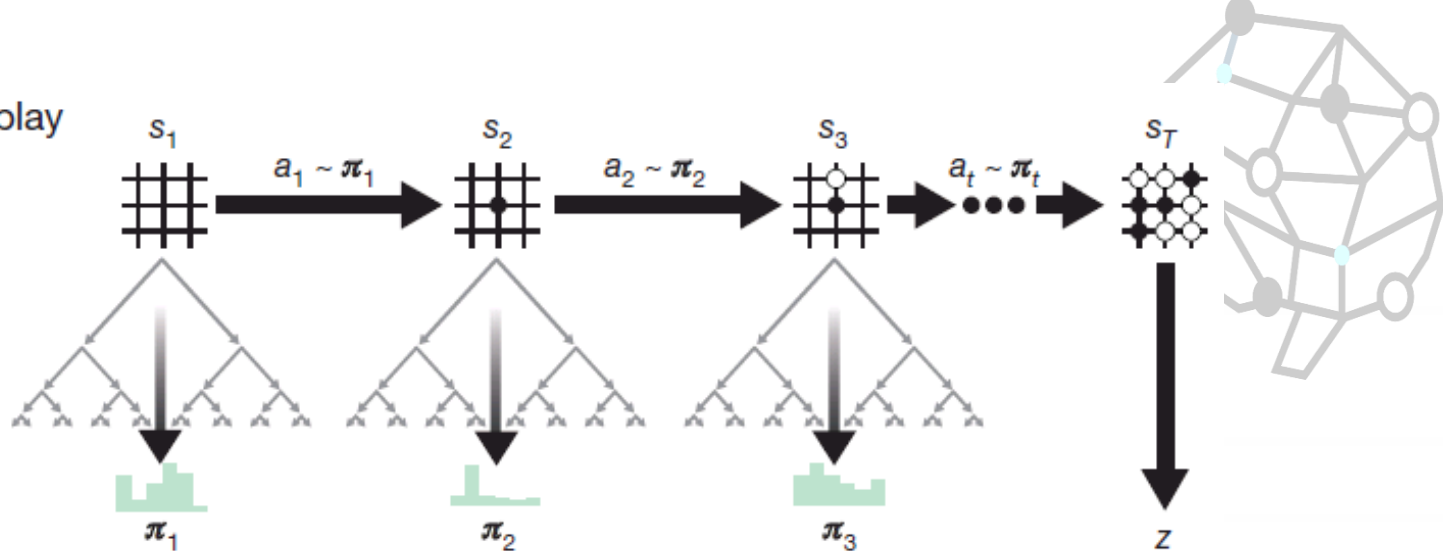
AlphaGo zero



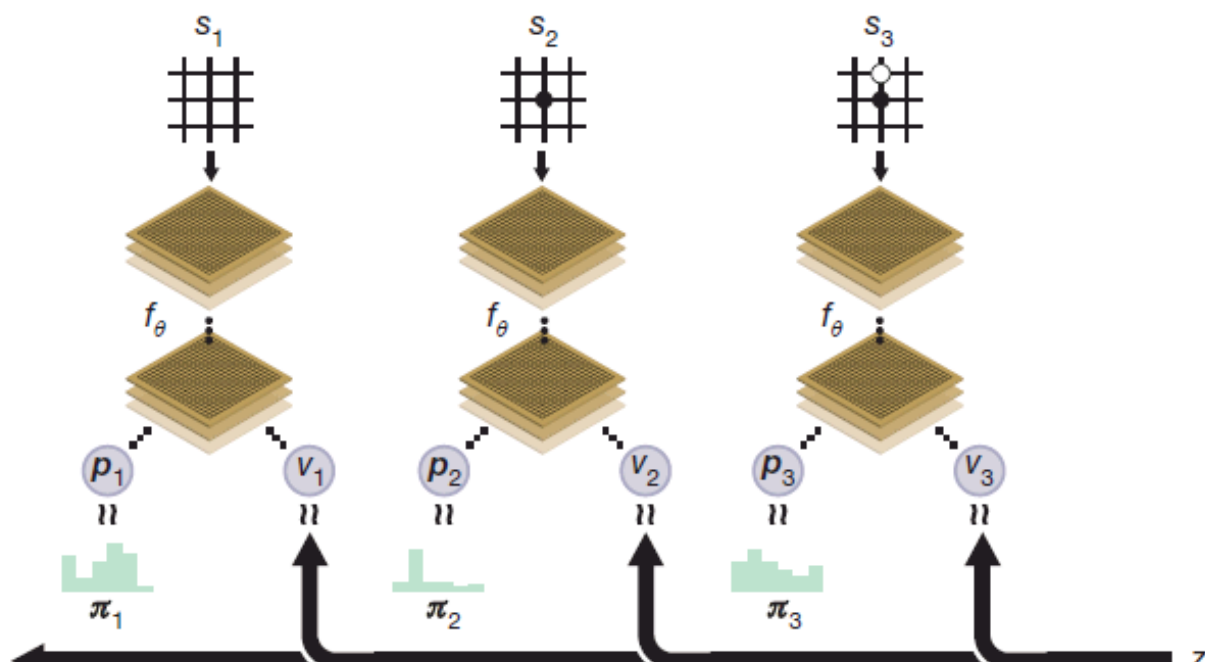
- Mastering the game of Go without human knowledge(David Silver, 2017/10)



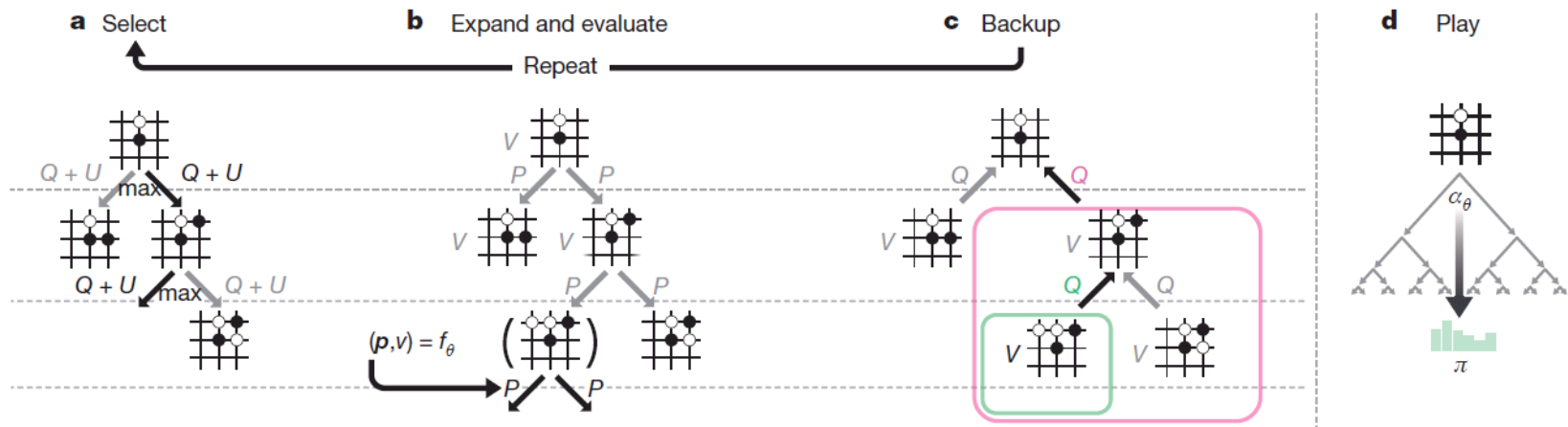
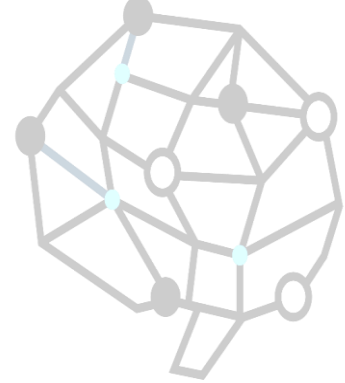
a Self-play



b Neural network training



MCTS in AlphaGo Zero



A blue-tinted background image showing a hand moving a chess piece on a chessboard. The text "THE END." is centered in white, with a white horizontal line underneath it.

THE END.
